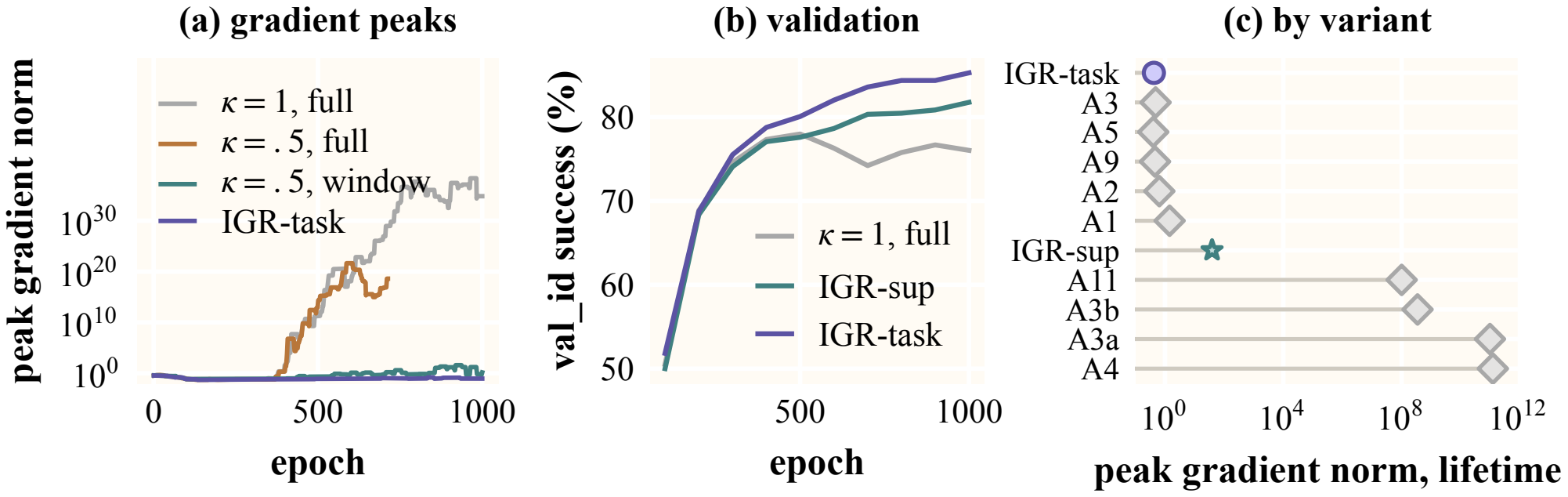
(a) gradient peaks
peak gradient norm
$\kappa = 1$, full
$\kappa = .5$, full
$\kappa = .5$, window
IGR-task
10^{30}
10^{20}
10^{10}
10^{0}
0
500
1000
epoch
(b) validation
val_id success (%)
80
70
60
50
$\kappa = 1$, full
IGR-sup
IGR-task
500
1000
epoch
(c) by variant
IGR-task
A3
A5
A9
A2
A1
IGR-sup
A11
A3b
A3a
A4
10^{0}
10^{4}
10^{8}
10^{12}
peak gradient norm, lifetime